
Personal Information

Full Name: Penghui Yao

Nationality: Chinese

Contact Information

Department of Computer Science and Technology, Nanjing University, 163 Xianlin Avenue, Qixia District, Nanjing, Jiangsu, 210023, China

Email: phyao1985@gmail.com

Employment

- 2024 - present: Professor
Department of Computer Science and Technology, Nanjing University
- 2018 - 2024: Associate professor
Department of Computer Science and Technology, Nanjing University
- 2016 - 2018: Hartree Postdoctoral fellow
Joint Center for Quantum Information and Computer Science, University of Maryland.
Supervisors: Andrew Childs and Jacob Taylor
- 2015 - 2016: Postdoctoral fellow
Institute for Quantum Computing, University of Waterloo
Supervisors: Debbie Leung and Ashwin Nayak
- 2014 - 2015: Postdoctoral fellow
Centrum Wiskunde & Informatica
Supervisor: Ronald de Wolf
- 2013 - 2014: Research associate
Centre for Quantum Technologies, National University of Singapore.
Supervisor: Rahul Jain

Education

- 2008 - 2013: Ph.D. in computer science
Centre for Quantum Technologies, National University of Singapore.
Supervisors: Rahul Jain and Miklos Santha
Thesis: Studies in communication complexity and semidefinite programs.

- 2006 - 2008: Graduate student in computer science
Institute of Software, Chinese Academy of Sciences
Supervisor: Angsheng Li
- 2003 - 2006: B.S. in mathematics
Department of Mathematics, East China Normal University
- 2002 - 2003
Department of Mathematics, Shanghai University of Electric Power

Professional activities

- Program Committee: QIP 2018, TQC 2021, TQC 2022, QIP 2023, QIP 2024, TQC 2024, AQIS 2024, ALT 2025, QIP 2025, AQIS 2025.

Publications

1. Optimal T Counts under Sparsity: from QROM to State Preparation and Block Encoding
[\[arXiv:2607.28260\]](#)
Tongyang Li, Fengning Ou, Xinzhao Wang, Penghui Yao, Pei Yuan, Shengyu Zhang
2. Dimension-Free Approximate Tensorization of Quantum Hypercontractivity for Qudit Depolarizing Semigroups [\[arXiv:2606.17729\]](#)
Yangjing Dong, Li Gao, Fengning Ou, Penghui Yao, Haigang Zhou
AQIS 2026
3. On the Computational Complexity of Geometrically Local QACO Circuits [\[arXiv:2604.07178\]](#)
Yangjing Dong, Fengning Ou, Penghui Yao
4. A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity [\[arXiv:2511.17227\]](#)
Xudong Wu, Guangxu Yang, Penghui Yao
ICALP 2026
5. Linear-Size QACO Channels: Learning, Testing and Hardness [\[arXiv:2510.00593\]](#)
Yangjing Dong, Fengning Ou, Penghui Yao
6. Nonlocal Games in the High-Noise Regime: Optimal Quantum Values and Rigidity
[\[arXiv:2509.20350\]](#)
Honghao Fu, Minglong Qin, Haochen Xu, Penghui Yao
7. Efficient Non-Adaptive Quantum Algorithms for Tolerant Junta Testing [\[arXiv:2508.17306\]](#)
Zongbo Bao, Yuxuan Liu, Penghui Yao, Zekun Ye, Jialin Zhang
SOSA 2026; QIP 2026
8. Optimal Quantum Sampling on Distributed Databases [\[arXiv:2506.07724\]](#)
Longyun Chen, Jingcheng Liu, Penghui Yao
SPAA 2025
9. Experimental Efficient Influence Sampling of Quantum Processes [\[arXiv:2506.07103\]](#)
*Hao Zhan, *Zongbo Bao, Zekun Ye, Qianyi Wang, Minghao Mi, #Penghui Yao, #Lijian Zhang
Authors listed by contribution; * equal contribution; # corresponding authors
10. Parallel Kac's Walk Generates PRU [\[arXiv:2504.14957\]](#)
Chuhan Lu, Minglong Qin, Fang Song, Penghui Yao, Mingnan Zhao
11. A Pseudorandom Generator for Functions of Low-Degree Polynomial Threshold Functions
[\[arXiv:2504.10904\]](#)
Penghui Yao, Mingnan Zhao
ICALP 2025

12. Distributed Quantum Simulation [[arXiv:2411.02881](#)]
Tianfeng Feng, Jue Xu, Wenjun Yu, Zekun Ye, #Penghui Yao, #Qi Zhao
Authors listed by contribution; # corresponding authors
13. On the Computational Power of QAC0 with Barely Superlinear Ancillae [[arXiv:2410.06499](#)]
Anurag Anshu, Yangjing Dong, Fengning Ou, Penghui Yao
QIP 2025; STOC 2025
14. Almost Optimal Algorithms for Token Collision in Anonymous Networks [[arXiv:2408.10519](#)]
Sirui Bai, Xinyu Fu, Xudong Wu, Penghui Yao, Chaodong Zheng
DISC 2024
15. The Computational Advantage of MIP* Vanishes in the Presence of Noise [[arXiv:2312.04360](#)]
Yangjing Dong, Honghao Fu, Anand Natarajan, Minglong Qin, Haochen Xu, Penghui Yao
CCC 2024; QIP 2025; Journal of the ACM 2025
16. Hypercontractivity for Quantum Erasure Channels via Variable Multipartite Log-Sobolev Inequality [[arXiv:2311.14321](#)]
Zongbo Bao, Yangjing Dong, Fengning Ou, Penghui Yao
IEEE Transactions on Information Theory 2025
17. Communication Complexity of Common Randomness Generation with Isotropic States [[arXiv:2311.04723](#)]
Yangjing Dong, Penghui Yao
IEEE Transactions on Information Theory 2024
18. A Cryptographic Perspective on the Verifiability of Quantum Advantage [[arXiv:2310.14464](#)]
Nai-Hui Chia, Honghao Fu, Fang Song, Penghui Yao
AQIS 2024; ACM Transactions on Quantum Computing 2025
19. The Generations of Classical Correlations via Quantum Schemes [[arXiv:2304.12690](#)]
Zhenyu Chen, Lijinzhi Lin, Xiaodie Lin, Zhaohui Wei, Penghui Yao
IEEE Transactions on Information Theory 2024
20. Quantum and Classical Communication Complexity of Permutation-Invariant Functions [[arXiv:2401.00454](#)]
Ziyi Guan, Yunqi Huang, Penghui Yao, Zekun Ye
STACS 2024; IEEE Transactions on Information Theory 2025
21. Quantum Pseudorandom Scramblers [[arXiv:2309.08941](#)]
Chuhan Lu, Minglong Qin, Fang Song, Penghui Yao, Mingnan Zhao
QIP 2024; TCC 2024
22. On the Fine-Grained Query Complexity of Symmetric Functions [[arXiv:2309.11279](#)]
Supartha Podder, Penghui Yao, Zekun Ye
ISAAC 2023; Computational Complexity 2025
23. On Testing and Learning Quantum Junta Channels [[arXiv:2305.12097](#)]
Zongbo Bao, Penghui Yao
COLT 2023; IEEE Transactions on Pattern Analysis and Machine Intelligence 2025
24. Decidability of Fully Quantum Nonlocal Games with Noisy Maximally Entangled States [[arXiv:2211.10613](#)]
Minglong Qin, Penghui Yao
QIP 2023; ICALP 2023; Algorithmica 2025
25. Quantum Complexity of Weighted Diameter and Radius in CONGEST Networks [[arXiv:2206.02767](#)]
Xudong Wu, Penghui Yao
PODC 2022; IEEE Transactions on Information Theory 2026
26. Polynomial-Time Approximation of Zero-Free Partition Functions [[arXiv:2201.12772](#)]
Penghui Yao, Yitong Yin, Xinyuan Zhang
ICALP 2022

27. On the Gaussian Surface Area of Spectrahedra [[arXiv:2112.01463](#)]
Srinivasan Arunachalam, Oded Regev, Penghui Yao
Geometric Aspects of Functional Analysis: Israel Seminar (GAFA) 2022
28. Nonlocal Games with Noisy Maximally Entangled States Are Decidable [[arXiv:2108.09140](#)]
Minglong Qin, Penghui Yao
SIAM Journal on Computing 2021
29. Complexity of Eccentricities and All-Pairs Shortest Paths in the Quantum CONGEST Model [[arXiv:2206.02766](#)]
Changsheng Wang, Xudong Wu, Penghui Yao
SPIN 2021
30. Quantum and Classical Hybrid Generations for Classical Correlations [[arXiv:2007.10673](#)]
Xiaodie Lin, Zhaohui Wei, Penghui Yao
IEEE Transactions on Information Theory 2022
31. Quantum Verification of NP Problems with Single Photons and Linear Optics [[arXiv:2008.05453](#)]
Aonan Zhang, Hao Zhan, Junjie Liao, Kaimin Zheng, Tao Jiang, Minghao Mi, #Penghui Yao, #Lijian Zhang
Authors listed by contribution; Light: Science & Applications 2021
32. Positive Spectrahedra: Invariance Principles and Pseudorandom Generators [[arXiv:2101.08141](#)]
Srinivasan Arunachalam, Penghui Yao
STOC 2022
33. On the Compression of Messages in the Multi-Party Setting [[arXiv:1808.06449](#)]
Anurag Anshu, Penghui Yao
IEEE Transactions on Information Theory 2020
34. A Doubly Exponential Upper Bound on Noisy EPR States for Binary Games [[arXiv:1904.08832](#)]
Penghui Yao
QIP 2020; superseded by arXiv:2108.09140
35. Quantum Insertion-Deletion Channels [[arXiv:1901.00984](#)]
Janet Leahy, Dave Touchette, Penghui Yao
36. Expected Communication Cost of Distributed Quantum Tasks [[arXiv:1605.04601v3](#)]
Anurag Anshu, Ankit Garg, Aram Harrow, Penghui Yao
IEEE Transactions on Information Theory 2018
37. Capacity Approaching Coding for Low Noise Interactive Quantum Communication, Part I: Large Alphabets [[arXiv:2001.02818](#)]
Debbie Leung, Ashwin Nayak, Ala Shayeghi, Dave Touchette, Penghui Yao, Nengkun Yu
IEEE Transactions on Information Theory 2021; extended abstract in STOC 2018
38. Exponential Separation of Quantum Communication and Classical Information [[arXiv:1611.08946](#)]
Anurag Anshu, Dave Touchette, Penghui Yao, Nengkun Yu
STOC 2017; QIP 2017 plenary talk
39. Multipartite Quantum Correlation and Communication Complexities [[arXiv:1405.6015](#)]
Rahul Jain, Zhaohui Wei, Penghui Yao, Shengyu Zhang
Computational Complexity 2016
40. New One Shot Quantum Protocols With Application to Communication Complexity [[arXiv:1404.1366](#)]
Anurag Anshu, Rahul Jain, Priyanka Mukhopadhyay, Ala Shayeghi, Penghui Yao
IEEE Transactions on Information Theory 2016
41. Parity Decision Tree Complexity and 4-Party Communication Complexity of XOR-Functions Are Polynomially Equivalent [[arXiv:1506.02936](#)]
Penghui Yao
Chicago Journal of Theoretical Computer Science 2016

42. Lower Bound on Expected Communication Cost of Quantum Huffman Coding
Anurag Anshu, Ankit Garg, Aram Harrow, Penghui Yao
TQC 2016
43. A Parallel Repetition Theorem for Entangled Two-Player One-Round Games under Product Distributions [[arXiv:1311.6309](#)]
Rahul Jain, Attila Pereszlényi, Penghui Yao
CCC 2014
44. A Direct Product Theorem for Two-Party Bounded-Round Public-Coin Communication Complexity [[arXiv:1201.1666](#)]
Rahul Jain, Attila Pereszlényi, Penghui Yao
FOCS 2012; special issue in Algorithmica
45. A Parallel Approximation Algorithm for Positive Semidefinite Programming [[arXiv:1104.2502](#)]
Rahul Jain, Penghui Yao
FOCS 2011
46. Adversary Lower Bounds for Nonadaptive Quantum Algorithms [[arXiv:0804.1440](#)]
Pascal Koiran, Jürgen Landes, Natacha Portier, Penghui Yao
WoLLIC 2008; special issue in Journal of Computer and System Sciences

Manuscript

1. Raz-McKenzie Simulation with the Inner Product Gadget [[ECCC TR17-010](#)]
Xiaodi Wu, Penghui Yao, Henry Yuen